EXAMPLE: 4-3-4 TRAJECTORY PLANNING.

A JOINT OF A ROBOT MANIPULATOR IS TO FOLLOW A 4-3-4 TRAJECTORY IN TERMS OF POSITION. DETERMINE THE TRAJECTORY EQUATIONS FOR THE FOLLOWING CONDITIONS:

$$\theta_1 = 30^\circ \quad \dot{\theta}_1 = 0^\circ/s \quad \ddot{\theta}_1 = 0^\circ/s^2 \quad \tau_{1i} = 0 \quad \tau_{1F} = 2s.$$
$$\theta_2 = 50^\circ \quad \tau_{2i} = 0 \quad \tau_{2F} = 4s$$
$$\theta_3 = 90^\circ \quad \tau_{3i} = 0 \quad \tau_{3F} = 2s$$
$$\theta_4 = 70^\circ \quad \dot{\theta}_4 = 0^\circ/s \quad \ddot{\theta}_4 = 0^\circ/s^2$$

SOLUTION:

SUBSTITUTE CONDITIONS INTO M MATRIX GIVEN IN THE NOTES. THEN SOLVE FOR COEFFICIENTS

$$C = M^{-1}\theta$$ (SEE MATLAB FILE PROVIDED).

THESE GIVE THE FOLLOW COEFFICIENTS:

$$a_0 = 30 \quad b_0 = 50 \quad c_0 = 90$$
$$a_1 = 0 \quad b_1 = 20.477 \quad c_1 = -13.81$$
$$a_2 = 0 \quad b_2 = 0.714 \quad c_2 = -9.286$$
$$a_3 = 4.881 \quad b_3 = -0.833 \quad c_3 = 9.643$$
$$a_4 = -1.191 \quad c_4 = -2.024$$

THE 3 SEGMENTS BECOME

$$\theta_1(t) = 30 + 4.881t^3 - 1.191t^4 \qquad 0 \leq t \leq 2$$

$$\theta_2(t) = 50 + 20.477t + 0.714t^2 - 0.833t^3 \qquad 0 < t \leq 4$$

$$\theta_3(t) = 90 - 13.81t - 9.286t^2 + 9.643t^3 - 2.024t^4 \qquad 0 < t \leq 2.$$